Theorem 8.7 (Bayes' Theorem)

Let $B_1, \ldots, Bn$ be n mutually exclusive events such that where S is the sample space of the random experiment. If $P(B_j) > 0$ for $j = 1, 2, \ldots, n$, then for any event A of the same experiment with $P(A) > 0$,

$$P(B_j/A) = \frac{P(A/B_j)P(B_j)}{P(A/B_1)P(B_1) + P(A/B_2)P(B_2) + \ldots + P(A/B_n)P(B_n)}, \; j = 1, 2, \ldots, n.$$

[This Theorem is due to Rev. Thomas Bayes (1701-1761), an English philosopher and a priest. This work was published posthumously by his friend Richard Price during 1763 in the name of Bayes.]

Proof:

For each event B_j, $i = 1, 2, \ldots, n$, by the definition of conditional probability

$$P(B_j/A) = \frac{P(B_j \cap A)}{P(A)}$$

$$= \frac{P(A/B_j)P(B_j)}{P(A)} \quad \text{for } j = 1, 2, \ldots, n$$

Then, by the generalization of Theorem 8.5,

$$P(B_j/A) = \frac{P(A/B_j)P(B_j)}{P(A/B_1)P(B_1) + P(A/B_2)P(B_2) + \ldots + P(A/B_n)P(B_n)}, \; j = 1, 2, \ldots, n.$$

Example 8.21

Mr. Arivazhagan, Mr. Ilavarasan and Mr. Anbarasan attended an interview conducted for appointing a Physical Teacher in a school. Mr. Arivazhagan has 45% chance for selection, Mr. Ilavarasan has 28% chance and Mr.Anbarasan has 27% chance. Also, the chance for implementing monthly Mass Drill (MD) programme in the school is 42% if Mr.Arivazhagan is appointed; 40% if Mr.Ilavarasan is appointed; and 48% if Mr.Anbarasan is appointed.

Find the probability that the mass drill is implemented by if:

(i) Mr.Arivazhagan is appointed as the Physical Education Teacher.

(ii) Mr.Ilavarasan is appointed as the Physical Education Teacher.

(iii) Mr.Anbarasan is appointed as the Physical Education Teacher.